What a global source bound tells us

Each crossing region contributes a finite polynomial

Let $a_N(q;\delta)$ be one coefficient at level N, with normalized face degrees q and real spectral parameter δ. Its upper rate is

$$N^{-1}\log|a_N(q;\delta)| \quad \leq \quad H_\delta(q) \;+\; o(1).$$

This holds at both parities and at boundary degrees, including zeros.

The face degrees must agree across the diagram

Each region supplies degree 1. Each ordinary face requires degree 1; the two marked faces require 0. Maximize the sum of local rates subject to these shared constraints, and call the maximum M_D.

$|S_N| \quad \leq \quad e^{N(M_D+o(1))}$ for the full shared numerator S_N.

Geometry restricts the possible maximizing configurations

On a diagram satisfying the geometric certificate, every ordinary incidence has positive load and all regions use common face weights. The marked zeros remain fixed. The possible local forms are:

Central cross sector

$\delta \;=\; 0$

Both middle loads are below $\frac{1}{2}$.

This is the complete geometric configuration.

One-sided sector

$0 \;<\; \delta \;<\; \frac{1}{2}$

One middle load exceeds $\frac{1}{2}$.

The two orientations are left and right.

Marked cap corner

$\delta \;=\; \frac{1}{2}$

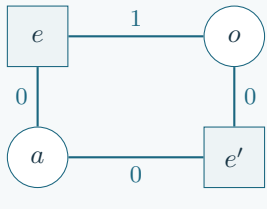
Two marked incidences have zero weight.

The other weights agree; at most two caps occur.

A non-central region forces a phase mismatch

In the four-step incidence cycle, 0 means equal phases and 1 means phases differing by π. An odd sum cannot be satisfied around the cycle.

This detects cancellation, but does not measure its size.



What remains for general diagrams

Let Φ_D denote the complete geometric exponential rate. The cancellation must be strong enough to bring the full source below that rate outside the geometric contribution. A positive loss relative to a larger upper bound does not establish this.

The seam boundary is controlled; a flat cap prevents a universal local exponential-loss argument. This route alone does not prove $M_D \quad \leq \quad \Phi_D$ for every certified plat.

Strict Conway trees already have a full volume theorem by a separate argument.

For a maximum constrained to non-central sectors with value at least Φ_D, the seam theorem supplies a uniform positive overload margin. Squares in the cycle are crossing regions; circles are shared faces.